

LETTER

Calculation and Conscience: Motivations for the Substantive Representation of Ethnic Minorities

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Abstract

Existing work shows that minority legislators are more likely to represent their group's interests compared to dominant group legislators. However, it is unclear whether this is due to intrinsic motivations or electoral incentives. We use a regression discontinuity design (RDD) to analyze ethnic minority representation in the UK Parliament. By comparing white MPs who narrowly beat minority candidates to minority MPs who narrowly beat white candidates, the RDD controls for electoral incentives since it holds constant constituency factors correlated with a minority parliamentary win. Analyzing over 1 million parliamentary questions and speeches, we find that minority MPs are more likely than white MPs to discuss issues important to ethnic minorities. Additional evidence suggests that narrowly elected MPs face similar electoral incentives and that minority MPs representing minorities face re-election penalties. Our findings are consistent with the idea that minority substantive representation is driven at least in part by intrinsic motivations.

Keywords: ethnic minority representation; substantive representation; intrinsic motivation; UK Parliament

Introduction

Over the past two decades, Western democracies have seen notable increases in ethnic minority legislators (Erzeel and Rashkova 2023). This has prompted extensive research into these legislators' roles and impacts, particularly regarding the relationship between descriptive and substantive representation following seminal works by Phillips (1998) and Pitkin (1967). Studies consistently show minority legislators are more likely than majority counterparts to provide constituency services to minorities (Lowande et al 2019), raise awareness of minority policy priorities (Saalfeld 2014), and support legislation advancing minority interests (Preuhs 2006) and rights (Grose 2005). However, as noted by Broockman (2013), Crisp et al (2018), and others, whether this positive association primarily stems from intrinsic motivation or electoral incentives remains unclear.

Increasing descriptive representation of minorities is considered key for reducing social and economic inequalities between dominant majorities and marginalized groups (Griffin 2014). Following Mansbridge (1999), we define descriptive representation as the extent to which representatives resemble their constituents in social or demographic identities. By intrinsic motivation, we mean representatives' genuine commitment to advancing the interests of groups with whom they share identity-based characteristics, independent of electoral calculations –

contrasting with strategic motivation to secure minority votes. Theory suggests minority politicians are more likely to promote minorities' interests – that is, substantively represent them – partly because shared experiences and social bonds create group solidarity (Williams 2000). Such intrinsic motivation helps explain why minority representatives more often promote minority group welfare (Whitby 2000).

However, the positive correlation between descriptive and substantive representation may instead reflect strategic electoral motivations. Since minority politicians often represent constituencies with large minority populations, their focus on minority interests may be a rational electoral strategy. Consistent with this, Saalfeld and Bischof (2013) found all British MPs respond to minority constituents where these populations are concentrated. Similarly, Crisp *et al* (2018) found that in New Zealand, Māori MPs are no more likely to represent their group than non-Māori MPs when institutional incentives for such behavior are absent. Bailer *et al* (2022) provide further evidence of electoral motivation, showing that as minority MPs gain seniority, their focus on group-specific representation declines, shifting towards broader national policy areas.¹

If minority MPs' actions reflect solely a rational response to electoral incentives, then perhaps descriptive representation is unnecessary for substantive representation (though it may still be important for symbolic reasons [Vishwanath, 2025]). Given the importance of establishing whether intrinsic motivations exist alongside electoral incentives for both theory and policy, it is no surprise that studies have attempted to demonstrate the existence of intrinsic motivations. Identifying such intrinsic motivations, however, is methodologically challenging because both explanations are generally observationally equivalent (Broockman 2013, 523–24). As we discuss below, past studies have struggled to overcome the problem of controlling for compositional effects of two types: the composition of places that elect minority representatives and the composition of characteristics of minority representatives.

Existing studies typically use regression frameworks that control for constituency characteristics when examining minority MP effects on substantive representation (for example, Canon 2020; Grose 2005; Juenke and Preuhs 2012). While Saalfeld (2011) improves this approach by adding an interaction between a minority representative indicator and the minority population share, the fundamental identification challenge remains: these methods assume no unobservable confounding between minority descriptive representation and constituency demands for minority representation (for example, unobserved MP political ambition or constituency mobilization capacity), despite substantial observable differences between constituency types. Matching estimators (for example, Kopkin 2017) also cannot address unobservable confounding.

An alternative and innovative approach by Crisp *et al* (2018) exploits variation in New Zealand's electoral systems, where Māori MPs face different levels of dependence on minority voter support. Yet this strategy cannot rule out that candidate selection and sorting across systems may correlate with underlying representational motivations.

Audit experiments address observational challenges by manipulating constituent identity, thereby controlling for constituent needs (for example, Lowande *et al* 2019). Broockman (2013) innovates by also varying constituency status, providing evidence for intrinsic motivation by showing black representatives are more responsive to black citizens even outside their constituencies. While valuable, audit experiments suffer from two limitations: first, they may not distinguish intrinsic motivation from statistical discrimination and in-group bias (see, for example, Costa 2017 on generalized legislator in-group favoritism), and second, constituent service findings may not generalize to legislative behavior where MPs face different electoral

¹See Broockman (2013, 522–23) for a discussion on how the strategic pursuit of re-election can explain why minority MPs are more likely to advance their group's interests.

and party pressures (see also Grossman and Slough 2022 on the limited scalability of constituent services).²

We complement the existing literature with a regression discontinuity design (RDD) of close elections between white and minority candidates in the 2010–24 UK general elections. By comparing minority MPs who narrowly win against white candidates to white MPs who narrowly win against minority candidates at the victory cut-off, we ensure both groups face constituencies with similar characteristics, overcoming compositional effects and controlling for electoral incentives to identify intrinsic motivations.³

The UK provides a valuable context to explore minority representation. As a parliamentary democracy with substantial immigration, the UK's ethnic minority population rose from 14 per cent to 18.3 per cent between 2011 and 2021, prompting parties to field more minority candidates (Sobolewska 2013). From 2010 to 2024, 1,616 ethnic minority candidates contested elections across the UK (Appendix A). Minority MPs increased from 4 per cent (22 MPs) in 2010 to 14 per cent (90 MPs) in 2024. Combined with comprehensive parliamentary records and competitive elections, this enables RDD analysis examining substantive representation.

Our analysis of 600,000+ speeches and 700,000+ parliamentary questions classified using large language models (LLMs) shows narrowly elected minority MPs are more likely than white MPs to discuss issues important to ethnic minorities. Specifically, minority MPs allocate 6–7 percentage points (pp) more of their speeches to ethnic minority problems and rights (a 230–360 per cent increase), and are 17–20 pp more likely to take pro-immigration positions when discussing immigration (a near-tripling). These effects are robust and statistically significant at conventional levels. Analyses of parliamentary questions show consistent positive patterns, though not statistically significant. Additional tests confirm both groups face similar electoral incentives, constituency demographics, MP backgrounds, and overall parliamentary participation. Crucially, descriptive evidence suggests minority MPs face re-election penalties for representing minority issues – meaning they engage in this representation despite electoral costs rather than because of electoral benefits – providing additional support for an intrinsic motivation interpretation.

This article makes two key contributions. While previous RDD studies examined minorities' representational outcomes (for example, Bhalotra et al 2014; Hopkins and McCabe 2012; Kogan et al 2021), ours is the first to systematically describe the inferential challenges in existing studies as arising from compositional effects – both of places that elect minorities (Saalfeld 2011) and candidate sorting (Crisp et al 2018) – and to use RDD to address these effects.

Second, we systematically measure minorities' substantive representation through comprehensive parliamentary questions and speeches, going beyond roll-call votes (Geese et al 2015; Vishwanath, 2025) that may not reflect constituent concerns due to strong UK party discipline (Slapin et al 2018; Slapin and Kirkland 2020). Unlike audit studies focused on individual constituent services (Broockman 2013), our outcomes measure group-level representation. Additionally, our LLM classification of both speeches and questions provides more comprehensive measurement than studies using questions alone (Saalfeld and Bischof 2013; Saalfeld 2011).

Data and Methods

Election Results, Candidate Ethnic Background, and Sample Selection

We assess representation in five UK Parliament sessions following the 2010, 2015, 2017, 2019, and 2024 general elections. Our sample includes constituencies where at least one minority candidate

²Although surveys of legislators directly measure motivations (Sobolewska et al 2018), they may suffer from systematic social desirability bias among minority representatives and still cannot account for electoral incentives due to compositional effects.

³The sections Compositional Effects and Electoral Incentives provides evidence for this claim.

stands against a white candidate, yielding 1,053 MP Parliament instances from 2010 to 2024, of which 264 (25 per cent) are represented by minority MPs. To construct this sample and compute the RDD independent and running variables (detailed further in section Estimation Method), we use data on UK general election results from the Electoral Commission⁴ and data on 2010–2019 candidates’ ethnic backgrounds collected by Zonszein and Grossman (2024). We classify 2024 candidates’ ethnic groups based on predicted nationalities from their names using *OnoGraph*.

Parliamentary Activities: Speeches and Questions for Written Answer

Parliamentary speeches and questions for written answer (PQs) effectively capture substantive representation (Bird 2005; Kroeber 2018; Saalfeld and Bischof 2013). These activities influence policy at scale by affecting constituents and other MPs (Blumenau 2021; Dietrich *et al* 2019), and better reveal MPs’ commitment to group-specific interests than divisions – roll-call votes – as they face fewer party constraints (Proksch and Slapin 2012).

However, speeches and PQs differ in their value as signals of intrinsic motivation. PQ submission is nearly unlimited, so MPs may submit PQs without spending much political capital. Existing studies analyze only PQs (for example, Bailer *et al* 2022; Saalfeld and Bischof 2013; Saalfeld 2011). Therefore, while we include PQs, we center our analysis on parliamentary speeches. Speaking opportunities are restricted and allocated by the Speaker (Slapin *et al* 2018), meaning MPs must compete for limited time. This scarcity makes speech choices particularly revealing of representational priorities, helping distinguish intrinsic motivation from credit-claiming.

We web-scraped MP speeches and PQs from *TheyWorkForYou* (Appendix D). Our analysis focuses on the first 22 months following each election – the maximum period between 2017 and 2019, ensuring equal weight across sessions. This yields 142,750 speeches and 148,092 PQs from 1,053 MP Parliament instances (Appendix B).

Parliamentary Activities: Defining Minority Interests

Following existing literature, we focus on two issues relevant to ethnic minority voters: the problems and rights of ethnic minorities in the UK (including discrimination, equal opportunities, and bias in policing) and immigration. We examine both issue salience and positions, breaking immigration into inclusionary and exclusionary positions to extend existing salience-focused studies (for example, Saalfeld and Bischof 2013). Appendix C presents examples.

These are the two issues covered by Saalfeld and Bischof (2013). Since then, ethnic minority population increased from 14 to 18.3 per cent and minority MPs more than quadrupled, making it valuable to revisit substantive representation. Moreover, ethnic minority voters prioritize anti-discrimination policy (Heath *et al* 2013) and are more supportive of immigration (Martin and Blinder 2021), underscoring these domains’ centrality.

Parliamentary Activities: Classification and Outcome Measures

To classify speeches and PQs into issues, we used OpenAI’s GPT-3.5 after creating a gold standard from almost 1,500 manually coded excerpts. Our team classified each excerpt into relevant issues and determined constituency references. We prompted GPT-3.5 with classification instructions and refined prompts to achieve high accuracy and F1 scores against hand-coded benchmarks. Immigration speeches were further classified into anti-immigration, pro-immigration, and neutral stances using similar methods. Appendix E provides further details

⁴See <https://commonslibrary.parliament.uk/research-briefings/CBP-8647/>.

on this classification process and performance metrics. This automated classification enabled large-scale analysis that would have been prohibitively costly with human coding alone and outperformed dictionary approaches and trained text classification models (see, for example, Mellon et al 2024; Rathje et al 2024). LLM classification captures nuanced discussion that keyword searches would miss while maintaining consistency across over 1 million parliamentary actions.

For each MP Parliament instance, we measure speech on relevant issues as the proportion of an MP's total speeches discussing each issue. This scaling accounts for differences in parliamentary activity due to expertise, seniority, or institutional positions (see for example, Blumenau and Damiani 2021). Similarly, we measure PQs as the count of issue-relevant questions divided by months not serving as government minister (since ministers cannot submit PQs). We extract ministerial data from Wikipedia infoboxes, which complements Cagé and Dewitte (2025)'s data. Following existing studies (for example, Saalfeld 2011), we do not scale PQs by total questions submitted since PQ submission is unlimited.

We acknowledge all MP actions can be strategic. However, different actions face different constraints: while votes face strong party whipping, speeches enjoy greater autonomy and capture individual MP positions even when rebelling against party lines (Slapin and Kirkland 2020). Speeches are also more costly signals than PQs due to restricted speaking opportunities. Though imperfect, speeches represent the most feasible observable measure of intrinsic motivation in parliamentary settings.

To validate our RD design, we measure MPs' overall parliamentary activity: speeches, written questions, oral questions, petitions, and division participation. These variables help rule out compositional effects by testing whether white and minority MPs differ in overall parliamentary productivity. This addresses both the general concern from observational studies that minority MPs may be systematically different, and Marshall (2024)'s specific RDD concern about compensating differentials – where marginally winning minority MPs might be superior in other characteristics. The section Compositional Effects finds no evidence of such differentials.

Estimation Method

Our challenge is distinguishing intrinsic motivation from electoral incentives. Minority MPs typically represent constituencies with larger minority populations, confounding MP ethnicity with electoral pressures. Using a sharp RDD addresses this by comparing minority MPs who narrowly won against white candidates to white MPs who narrowly won against minority candidates at the victory cut-off. Close elections quasi-randomly assign MP ethnicity while holding constant constituency characteristics. Narrow margins are crucial because, on average, constituencies in landslide elections differ substantially, while those in competitive elections are remarkably similar (Appendix F.2). By holding constant constituency factors, the RDD isolates MP ethnicity – revealing intrinsic motivations.

We estimate the following weighted linear regression:

$$Y_{it} = \alpha + \beta_1 \text{VictoryMargin}_{it} + \tau \text{EthnicMinorityVictory}_{it} + \beta_2 (\text{VictoryMargin}_{it} \times \text{EthnicMinorityVictory}_{it}) + \beta' X_{it} + \epsilon_{it} \quad (1)$$

where Y_{it} is MP i 's parliamentary action in Parliament t . $\text{VictoryMargin}_{it}$ is the running variable measuring the ethnic minority candidate's winning margin. Following existing multiparty studies (for example, Klačnja and Titunik 2017), we define this as the vote share of the strongest ethnic minority candidate minus the strongest white opponent's vote share. This excludes constituencies with only white or minority candidates (sample characteristics in Appendix F.3). $\text{EthnicMinorityVictory}_{it}$ is a binary treatment indicating minority candidate victory

Table 1. Summary statistics

	Mean	SD	Min	Max	<i>N</i>
% Immigration speech	0.08	0.09	0.00	0.61	1,053
% Discrimination speech	0.07	0.09	0.00	0.83	1,053
% Pro-immigration speech	0.24	0.27	0.00	1.00	901
% Neutral/anti-immigration speech	0.75	0.27	0.00	1.00	901
Num. immigration PQs	0.41	1.06	0.00	17.21	1,053
Num. discrimination PQs	0.21	0.48	0.00	5.83	1,053
Minority victory margin	−19.86	33.04	−82.05	80.64	1,053
Minority win	0.25	0.43	0.00	1.00	1,053
MP: incumbent	0.66	0.47	0.00	1.00	1,053
MP: Labour	0.52	0.50	0.00	1.00	1,053
MP: Conservative	0.34	0.47	0.00	1.00	1,053
MP: Labour & minority	0.14	0.35	0.00	1.00	1,053
MP: Conservative & minority	0.08	0.28	0.00	1.00	1,053
% Ethnic minority	22.01	19.63	1.00	79.90	1,053
Pop. density	31.70	33.08	0.11	197.58	1,053
% Employed	59.93	6.11	40.90	74.60	1,053
% High econ deprivation	5.08	2.33	1.20	14.16	1,053
% Social housing	19.14	8.40	5.30	50.60	1,053

Note: *N* = MP Parliament instances. Speech measures are share of MP total speeches; PQs are share of non-ministerial months.

($I(\text{VictoryMargin}_{it} > 0) = 1$). We include predetermined covariates X_{it} (detailed in Appendix F.1) to reduce residual variation and standard errors. All tables report estimates with and without controls using robust bias-corrected inferences.

We estimate this equation using local polynomial methods via the `rdrobust` package, which fits linear regressions above and below the victory cut-off within an optimal bandwidth. This optimal bandwidth defines ‘narrow elections’ and is selected with a data-driven algorithm minimizing the mean squared error (MSE) of the RD estimate (Calonico et al 2014). Observations are weighted by their distance from the cut-off using a triangular kernel, with observations outside the optimal bandwidth receiving zero weight. As a result, the effective number of observations used for estimation varies across outcomes depending on the bandwidth selected, ranging from 136 to 258 MP Parliament instances in our main specifications.

Assuming continuity of potential outcomes at the cut-off, $\hat{\tau}$ captures the average treatment effect of descriptive representation on substantive representation of ethnic minorities at the cut-off. The section Compositional Effects and Appendix J.2 show compensating differentials are not a major concern. Table 1 presents summary statistics for outcomes, treatment, and controls.

Results: Substantive Representation

Figure 1 presents regression discontinuity estimates of electing an ethnic minority MP. The top panels examine issue salience (proportion of speeches devoted to specific topics), while the bottom panels examine positions on immigration. As a constituency moves from barely represented by a white MP to barely represented by an ethnic minority MP, we observe increases at the victory cut-off in attention to minority issues. In each panel, the two lines represent local linear regressions fitted separately on each side of the victory cut-off; the RD effect estimates reported in Table 2 measure the vertical distance between the points where these two lines meet the cut-off.

Table 2 quantifies these effects: narrowly elected minority MPs allocate 5.7–7.3 pp more of their speeches to ethnic minority problems and rights (columns 1–2, Panel A) – a 229–363 per cent increase, statistically significant at the 1 per cent level. Immigration speeches show positive (2.7–3.3 pp) but not statistically significant effects (columns 3–4, Panel A).

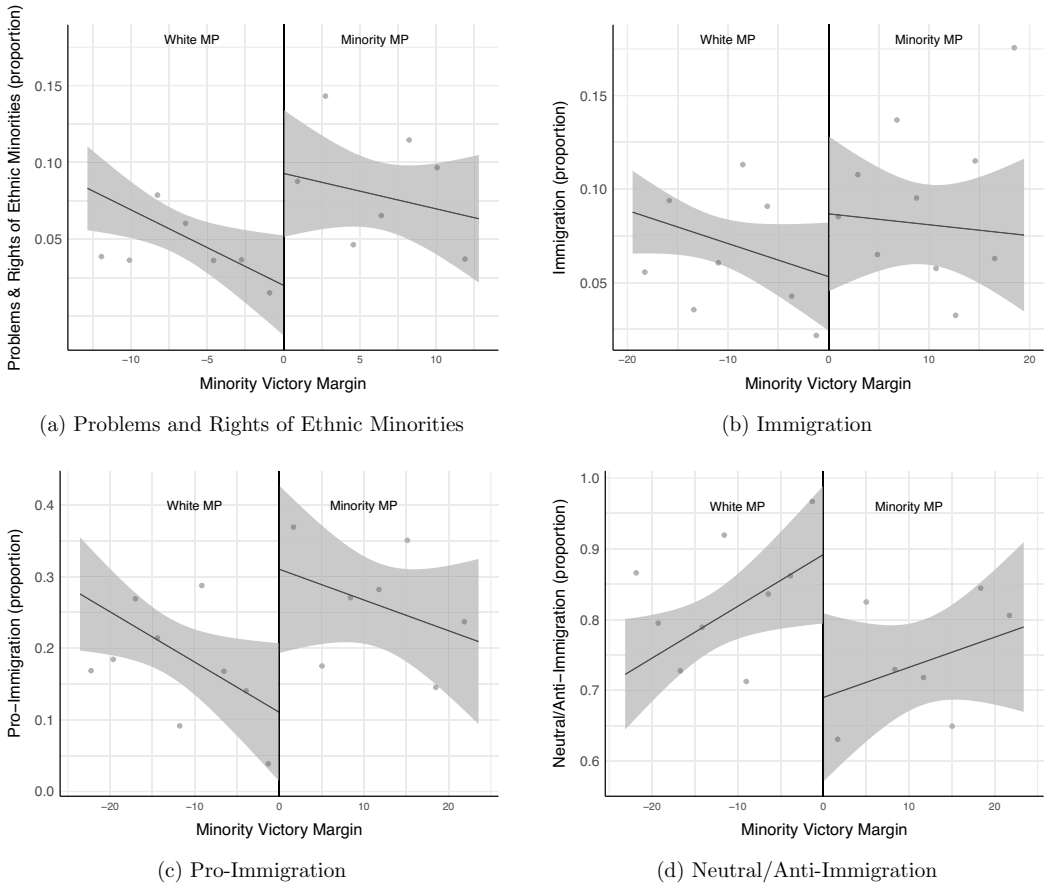


Figure 1. Substantive representation in parliamentary speeches: issue salience and positions.

Note: lines represent the average outcome (with 95 per cent confidence intervals) from local linear regression fitted to the sample of constituencies with minority victory margins within the MSE-optimal symmetric bandwidth around the victory cut-off. Points are the average outcome for equally spaced mimicking-variance bins.

White and minority MPs, however, differ in positions taken on immigration issues – consistent with minority voters’ pro-immigration preferences (Martin and Blinder 2021) (Appendix G). Minority MPs are 16.8–20 pp more likely to take pro-immigration positions (Panel B, columns 1–2). Given white MPs’ 11 per cent rate, minority MPs are nearly three times as likely to voice inclusionary positions. Appendix H shows consistent positive effects for parliamentary questions though not statistically significant, and Appendix I examines other issues.

Validity Tests and Robustness Checks

RD estimates require that candidates cannot manipulate election results around the cut-off. We conduct density tests (showing similar observation counts above and below the cut-off) and covariate tests (showing null RD effects on predetermined variables). Additional robustness checks include sensitivity tests for bandwidth choice and polynomial order, placebo cut-offs, and RD estimates with Armstrong and Kolesár (2020)’s bias-aware inference, and also under local randomization. Appendix J presents these tests, all of which support the validity of our RD design.

Our substantive representation results are not driven by party differences for three reasons. First, our model controls for MP party. Second, there is no discontinuity in MPs’ parties at the cut-

Table 2. Substantive representation: parliamentary speeches

	(1)	(2)	(3)	(4)
Panel A: Issue salience				
	Problems and rights of ethnic minorities in the UK		Immigration	
$I(\text{VictoryMargin}_{it} > 0) = 1$	0.073*** [0.024, 0.145] (0.027)	0.057*** [0.02, 0.111] (0.02)	0.033 [−0.015, 0.101] (0.026)	0.027 [−0.015, 0.079] (0.021)
Mean control	0.02	0.025	0.053	0.056
Percentage change	362.5	228.9	61.8	48
R^2	0.095	0.287	0.016	0.176
Eff. N (within bandwidth)	136	146	232	257
N	1,053	1,053	1,053	1,053
MSE-opt bw	12.873	13.999	19.587	21.588
Controls	N	Y	N	Y
Panel B: Issue positions				
	Pro-immigration		Neutral/Anti-immigration	
$I(\text{VictoryMargin}_{it} > 0) = 1$	0.2** [0.046, 0.392] (0.076)	0.168** [0.019, 0.333] (0.068)	−0.2** [−0.393, −0.046] (0.076)	−0.169** [−0.336, −0.02] (0.069)
Mean control	0.111	0.116	0.89	0.885
Percentage change	180.8	143.9	−22.5	−19.1
R^2	0.049	0.12	0.049	0.121
Eff. N (within bandwidth)	242	258	239	255
N	901	901	901	901
MSE-opt bw	23.565	24.567	23.5	24.394
Controls	N	Y	N	Y

Note: dependent variable is speeches on issue divided by total MP speeches (Panel A), or pro/anti-immigration speeches divided by total immigration speeches (Panel B). $I(\text{VictoryMargin}_{it} > 0) = 1$ is treatment effect at cut-off using local linear regression with triangular kernel and MSE-optimal bandwidth. Brackets show robust bias-corrected 95% CI. Standard errors in parentheses. Mean control: average for narrowly elected white MPs. Percentage change: effect as percentage of white MP mean. Eff. N (within bandwidth): observations within bandwidth; N : total observations. MSE-opt bw: optimal bandwidth size. Controls: constituency ethnic minority share, employment, deprivation, social housing, density; MP incumbency and party. Panel B excludes MPs with no immigration speeches. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

off (Appendix Figure J.2). Third, RD estimates are similar across Conservatives and left-leaning parties (Appendix K).⁵ While constrained by limited statistical power, this finding suggests minority MP effects reflect more than partisan differences.

Discussion: Minority MPs’ Intrinsic Motivation

The section Results: Substantive Representation shows increased ethnic minority descriptive representation has increased substantive representation. To interpret this as intrinsic motivation, we must rule out: (1) positive selection whereby minority MPs are more productive overall, and (2) differences in electoral incentives. Under intrinsic motivation, we expect null results – no differences in productivity or electoral incentives at the cut-off.

Compositional Effects

We test for positive selection by comparing parliamentary engagement across five metrics. If narrowly elected minority MPs were systematically more productive – a compensating differential

⁵Though Labour traditionally supports ethnic minorities and minority left-leaning MPs report stronger motivation to represent minorities (for example, Sobolewska et al 2018), this may not hold among narrowly elected MPs.

Table 3. MP parliamentary participation

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A						
	Petitions		Divisions		Speeches	
I(VictoryMargin _{it} > 0) = 1	0.003 [−0.003, 0.01] (0.003)	0.003 [−0.003, 0.01] (0.003)	−0.062 [−0.172, 0.026] (0.043)	−0.024 [−0.089, 0.051] (0.03)	−0.003 [−0.203, 0.163] (0.076)	−0.079 [−0.267, 0.078] (0.068)
Mean control	0.007	0.007	0.802	0.804	0.278	0.282
R ²	0.01	0.049	0.024	0.377	0.035	0.333
Eff. N (within bandwidth)	254	233	193	152	131	144
N	1,053	1,053	1,053	1,053	1,053	1,053
MSE-opt bw	21.352	19.729	17.25	14.369	12.454	13.826
Controls	N	Y	N	Y	N	Y
Panel B						
	Oral questions			Written questions		
I(VictoryMargin _{it} > 0) = 1	−0.005 [−0.033, 0.023] (0.012)	−0.009 [−0.038, 0.017] (0.011)	−0.313 [−9.049, 7.111] (3.375)	−2.918 [−11.426, 4.739] (3.377)		
Mean control	0.031	0.031	7.286	7.82		
R ²	0.056	0.274	0.007	0.093		
Eff. N (within bandwidth)	137	134	310	250		
N	1,053	1,053	1,053	1,053		
MSE-opt bw	13.017	12.81	25.118	21.053		
Controls	N	Y	N	Y		

Note: dependent variables are MP activities per parliamentary day (petitions, speeches, oral questions), division attendance rate, and written questions per month not serving as minister. $I(\text{VictoryMargin}_{it} > 0) = 1$ is treatment effect at cut-off using local linear regression with triangular kernel and MSE-optimal bandwidth. Brackets show robust bias-corrected 95% CI. Standard errors in parentheses. Mean control: average for narrowly elected white MPs. Eff. N (within bandwidth): observations within bandwidth; N total observations. MSE-opt bw: optimal bandwidth size. Controls: constituency ethnic minority share, employment, deprivation, social housing, density; MP incumbency and party. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

(Marshall 2024) – we would observe discontinuities. Instead, we expect null effects. Table 3 shows narrowly elected minority and white MPs have similar parliamentary engagement – estimates are substantively small and statistically insignificant. Appendix J.2 confirms narrowly elected minority and white MPs are similar in partisan affiliation, incumbency, education, political background, and demographic characteristics, and represent constituencies with similar socio-demographic profiles. These results suggest our RDD mitigates compositional effects and avoids bias from compensating differentials (Marshall 2024). However, MPs could still differ on unobserved characteristics, making RD estimates the bundled effect of ethnicity and these correlated characteristics.

Electoral Incentives

We test whether narrowly elected MPs face different electoral incentives using three measures: constituency ethnic minority population share (2011, 2021 Census), minority turnout, and minority vote share contribution (British Election Study).⁶ Under intrinsic motivation, we expect null effects; differences would suggest strategic responses.

Table 4 shows no discontinuities across all measures (point estimates are substantively small and statistically insignificant), confirming narrowly elected MPs face similar electoral incentives. This is consistent with an intrinsic motivation interpretation of our substantive representation findings.

Minority MPs might still strategically prioritize minority voters if this ensures re-election (Broockman 2013). However, within the optimal bandwidth, ethnic minorities comprise 23 per cent of the population while winners need 45–47 per cent of votes – making minority support

⁶The latter two measures are contemporaneous, so RD estimates provide descriptive rather than causal evidence.

Table 4. Electoral incentives: ethnic composition and voting behavior

	Minority population		Minority turnout		Minority contribution	
	(1)	(2)	(3)	(4)	(5)	(6)
I(VictoryMargin > 0) = 1	6.973 [−3.672, 21.888] (5.766)	−1.048 [−7.968, 6.496] (3.142)	0.037 [−0.27, 0.301] (0.121)	0.034 [−0.275, 0.285] (0.12)	−0.002 [−0.069, 0.09] (0.035)	−0.022 [−0.075, 0.03] (0.023)
Mean control	16.62	17.869	0.744	0.745	0.044	0.078
R ²	0.014	0.551	0.019	0.038	0.03	0.674
Eff. N (within bandwidth)	163	274	163	165	153	236
N	1,053	1,053	796	796	791	791
MSE-opt bw	15.295	23.19	19.304	19.486	18.662	27.16
Controls	N	Y	N	Y	N	Y

Note: dependent variables are constituency ethnic minority population share (columns 1–2), minority turnout (columns 3–4), and minority vote contribution to winner (columns 5–6). I(VictoryMargin_{it} > 0) = 1 is treatment effect at cut-off using local linear regression with triangular kernel and MSE-optimal bandwidth. Brackets show robust bias-corrected 95% CI. Standard errors in parentheses. Mean control: average for narrowly elected white MPs. Eff. N (within bandwidth): observations within bandwidth; N total observations. MSE-opt bw: optimal bandwidth size. Controls: constituency ethnic minority share, employment, deprivation, social housing, density; MP incumbency and party. ****p* < 0.01; ***p* < 0.05; **p* < 0.1.

insufficient for victory even with bloc voting. This differs from US majority–minority districts. Conversely, narrowly elected white MPs also need minority voters, as their co-partisan white electorate (~38 per cent) is insufficient. Therefore, differences may reflect, in part, minority MPs’ intrinsic motivations.

Appendix L provides descriptive evidence consistent with this interpretation. Among narrowly elected MPs, minority MPs who speak more frequently about ethnic minority rights or are pro-immigration have substantially lower re-election rates (38 vs. 70 per cent for minority rights; 39 vs. 58 per cent for pro-immigration), while white MPs face no such penalties for discussing these issues (26 vs. 30 per cent; 27 vs. 30 per cent). This asymmetric pattern suggests minority MPs prioritize minority representation despite – not because of – electoral consequences, further supporting that their behavior is not purely strategic vote-seeking.

Conclusion

This study provides causal evidence that ethnic minorities’ accession to parliament increases substantive representation. Ethnic minority MPs focus disproportionately on issues minority voters prioritize: discrimination, equal opportunities, and inclusionary immigration policy.

Additional evidence shows our RD design controls for electoral incentives while minority MPs face electoral penalties, suggesting intrinsic motivation is a driver of their behavior. This complements attitudinal research showing minority candidates feel greater responsibility to represent ethnic minorities (Sobolewska et al 2018), using behavioral evidence from a causal design.

This study has limitations. First, sample size constraints limit statistical power for heterogeneity analyses by ethnic group, party, or region, as only a subset of minority candidates compete in races close enough for RDD analysis. Second, our measures rely on observable parliamentary actions (speeches and questions), which inevitably contain strategic elements alongside intrinsic motivations. While speeches are more costly signals than questions due to restricted speaking opportunities, no measure perfectly captures unobservable motivations. Third, RDD identifies local average treatment effects at competitive election thresholds. Finally, as a single-country study, findings likely generalize best to democracies with similar electoral systems (first-past-the-post) and immigration histories, though further research across diverse contexts would strengthen these conclusions.

Our findings suggest descriptive representation delivers substantive democratic benefits beyond symbolic value. As democracies diversify, understanding these dynamics becomes crucial for institutional design promoting both inclusion and effective governance.

Supplementary material. Supplementary material for this article can be found at <https://doi.org/10.1017/S0007123426101690>.

Data availability statement. Replication data and code for this paper can be found at <https://doi.org/10.7910/DVN/ZCMRZM>.

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